

Air University



Organization and Assembly (Lab)

Year 2025

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Academic Year: 2024 – 2028

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BS – Cyber Security – Fall – 24 (A)

Date of Submission: October 07, 2025

Lab Task # 03

Task 1:

Write a program that will display a string on an output screen, in a COM style program.

Code

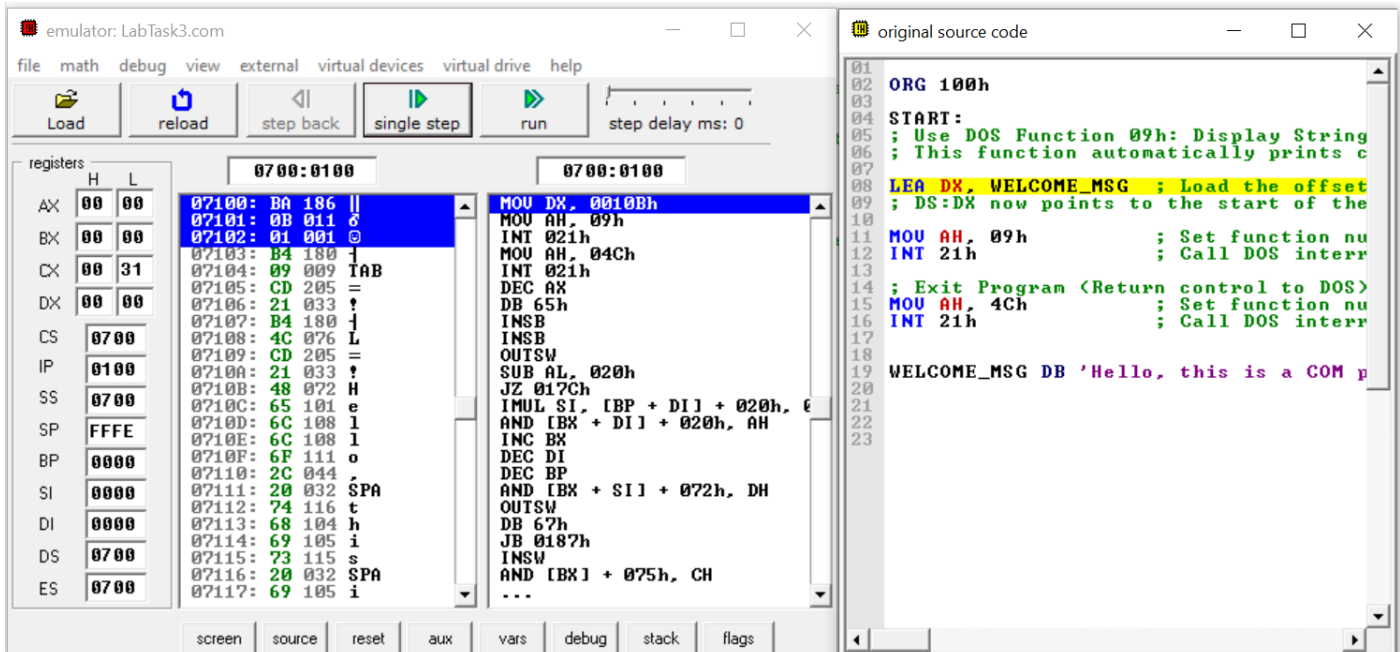
```

01
02 ORG 100h
03
04 START:
05 ; Use DOS Function 09h: Display String
06 ; This function automatically prints characters until it finds the '$' terminator.
07
08 LEA DX, WELCOME_MSG ; Load the offset address of WELCOME_MSG into DX.
09 ; DS:DX now points to the start of the string.
10
11 MOV AH, 09h ; Set function number for Display String
12 INT 21h ; Call DOS interrupt service
13
14 ; Exit Program (Return control to DOS)
15 MOV AH, 4Ch ; Set function number for Terminate Program
16 INT 21h ; Call DOS interrupt service
17
18
19 WELCOME_MSG DB 'Hello, this is a COM program output$', 0Dh, 0Ah
20

```

Output

- Execute the instructions at line 08 and loaded the address into DX using LEA.



- Execute the instructions at line 11 to set up the high byte of **AX** to **09h**.

The screenshot shows the LabTask3.com emulator interface. The registers window on the left displays the state of various registers. The source code window on the right shows the assembly code being executed.

Register	H	L
AX	00	00
BX	00	00
CX	00	31
DX	01	0B
CS	0700	
IP	0103	
SS	0700	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0700	
ES	0700	

The source code window shows the following assembly code:

```

01  ORG 100h
02
03  START:
04  ; Use DOS Function 09h: Display String
05  ; This function automatically prints c
06
07  LEA DX, WELCOME_MSG ; Load the offset
08  ; DS:DX now points to the start of the
09
10  MOV AH, 09h ; Set function nu
11  INT 21h ; Call DOS interr
12
13  ; Exit Program (Return control to DOS)
14  MOV AH, 4Ch ; Set function nu
15  INT 21h ; Call DOS interr
16
17  WELCOME_MSG DB 'Hello, this is a COM p
18
19
20
21
22
23

```

- Execute the instructions at line 12.

The screenshot shows the LabTask3.com emulator interface. The registers window on the left displays the state of various registers. The source code window on the right shows the assembly code being executed.

Register	H	L
AX	09	00
BX	00	00
CX	00	31
DX	01	0B
CS	0700	
IP	0105	
SS	0700	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0700	
ES	0700	

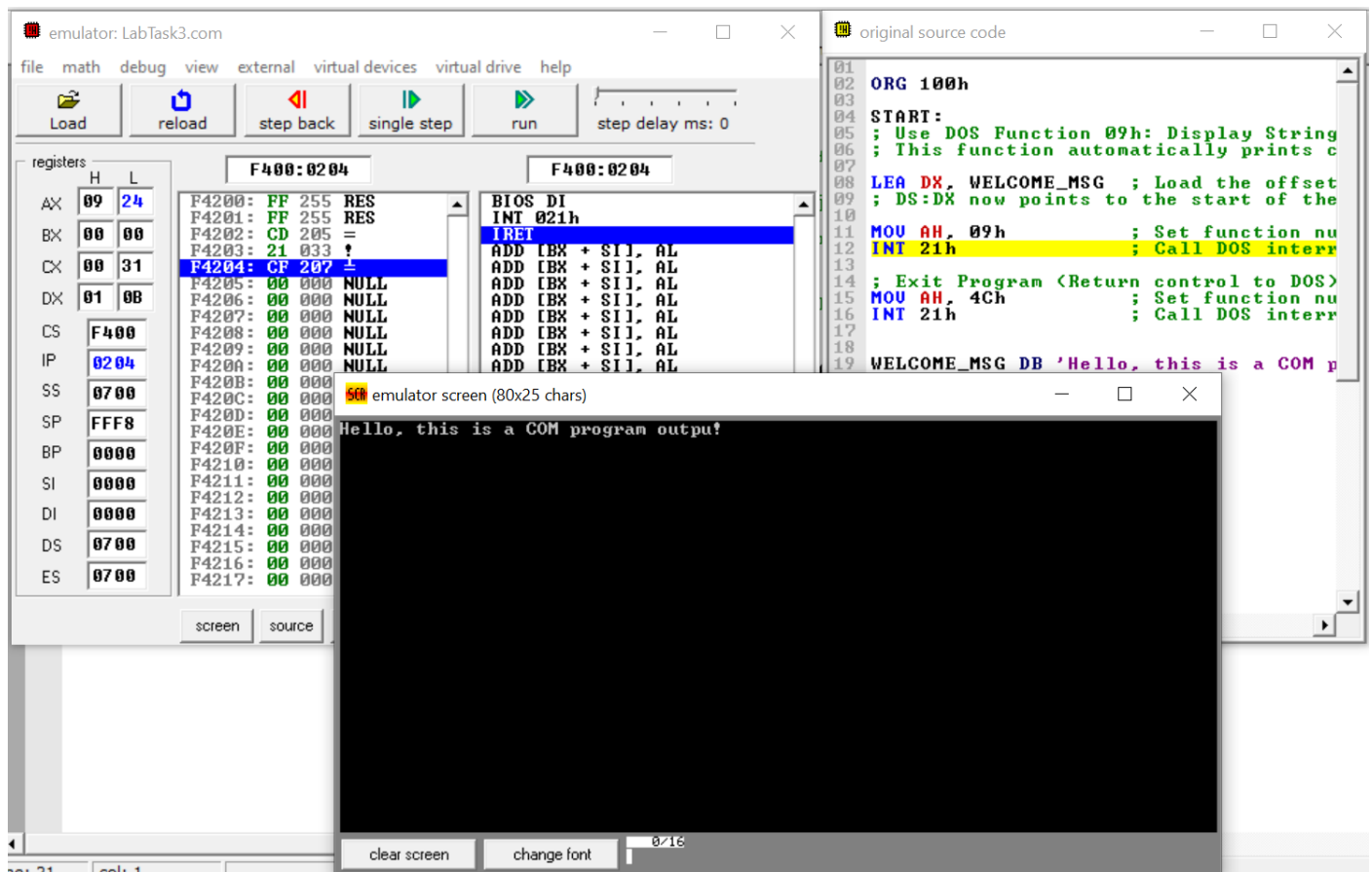
The source code window shows the following assembly code:

```

01  ORG 100h
02
03  START:
04  ; Use DOS Function 09h: Display String
05  ; This function automatically prints c
06
07  LEA DX, WELCOME_MSG ; Load the offset
08  ; DS:DX now points to the start of the
09
10  MOV AH, 09h ; Set function nu
11  INT 21h ; Call DOS interr
12
13  ; Exit Program (Return control to DOS)
14  MOV AH, 4Ch ; Set function nu
15  INT 21h ; Call DOS interr
16
17  WELCOME_MSG DB 'Hello, this is a COM p
18
19
20
21
22
23

```

- Final Output has appeared.



Task 2:

Write a program that will display a string on an output screen, in a EXE style program.

Code

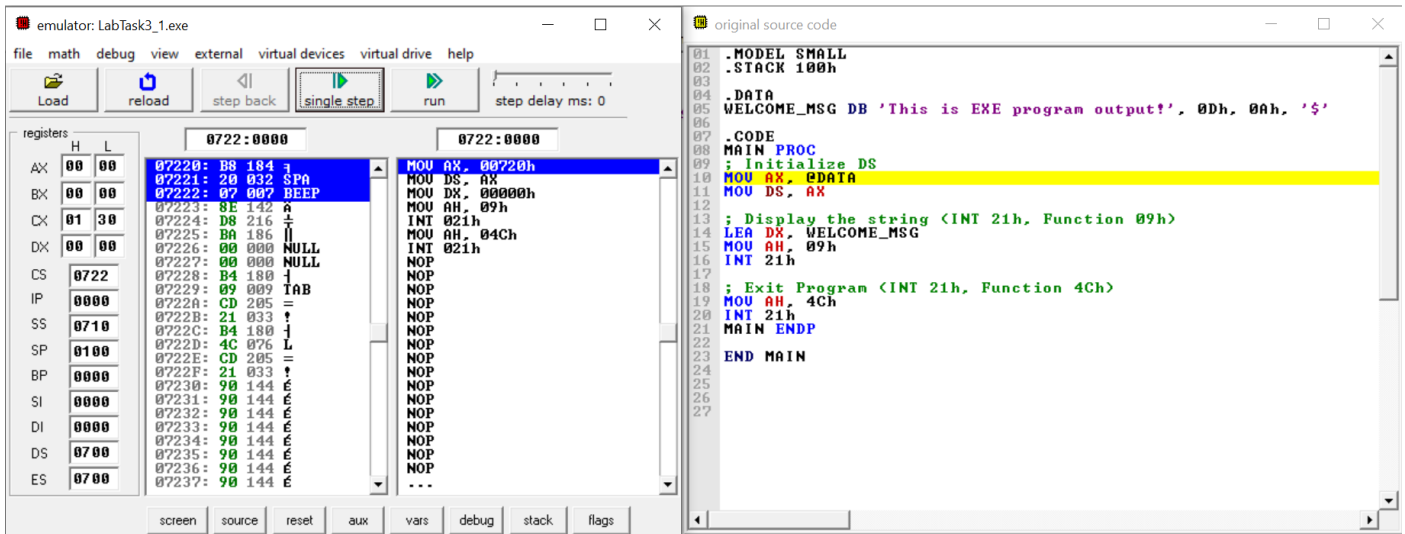
```

01 .MODEL SMALL
02 .STACK 100h
03
04 .DATA
05     WELCOME_MSG DB 'This is EXE program output!', 0Dh, 0Ah, '$'
06
07 .CODE
08 MAIN PROC
09     ; Initialize DS
10     MOV AX, @DATA
11     MOV DS, AX
12
13     ; Display the string (INT 21h, Function 09h)
14     LEA DX, WELCOME_MSG
15     MOV AH, 09h
16     INT 21h
17
18     ; Exit Program (INT 21h, Function 4Ch)
19     MOV AH, 4Ch
20     INT 21h
21 MAIN ENDP
22
23 END MAIN

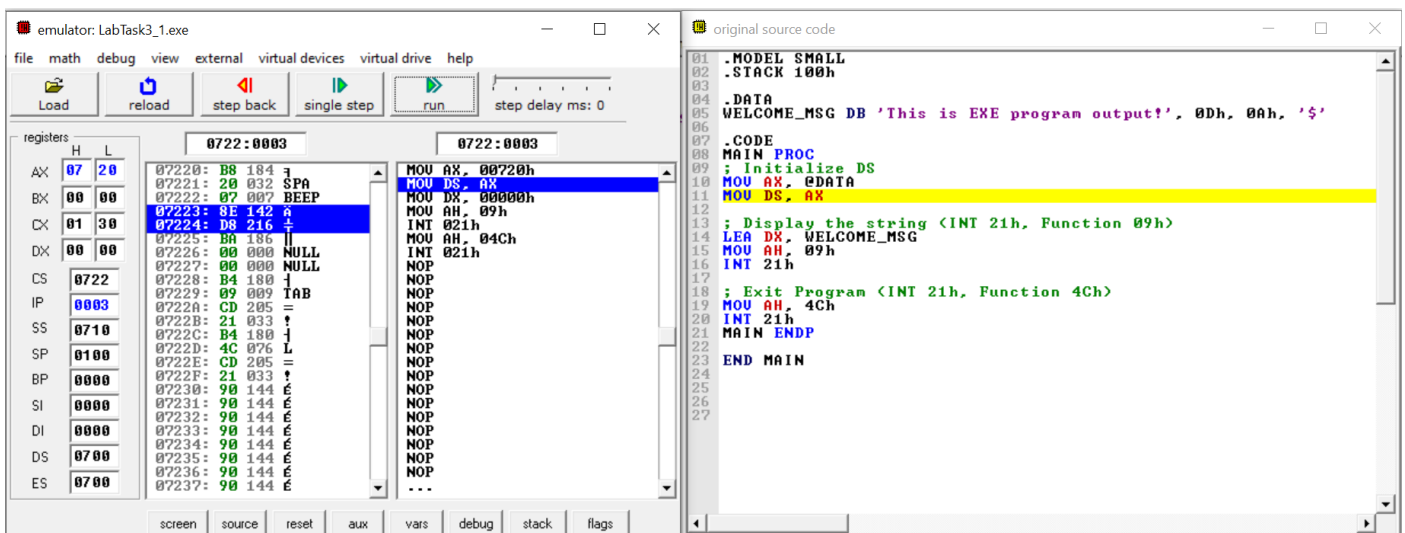
```

Output

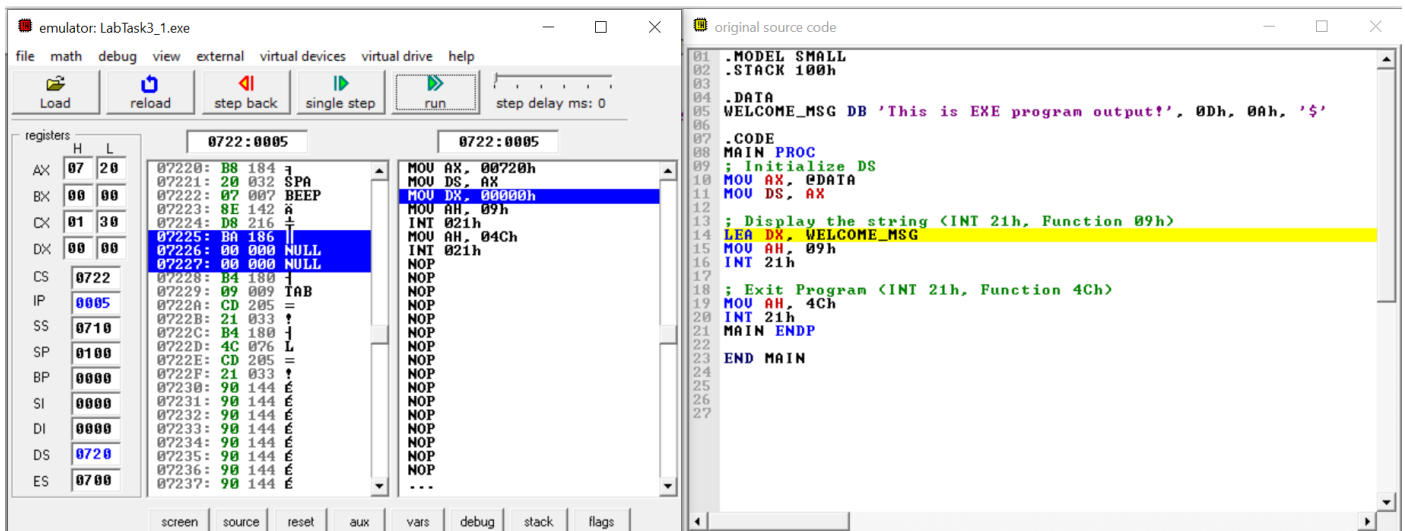
- Executed the instructions at line 11 and moved the @DATA into **AX** register.



- Executed instructions at line 12 to move **AX** into **DX**.



- Executed the instructions at line 14 to move the address of **WELCOME_MSG** into **DX** using **LEA** (Load Effective Address).



- The screenshot displays the LabTask3_1.exe emulator interface. The top menu bar includes options like file, math, debug, view, external, virtual devices, virtual drive, and help. Below the menu is a toolbar with buttons for Load, reload, step back, single step, and run, along with a step delay slider set to 0 ms.

The main window is divided into two panes. The left pane shows the registers (AX, BX, CX, DX, SI, DI, SP, BP, SI, DI, DS, ES) and their values. The right pane shows the assembly code being executed, with the current instruction highlighted in blue.

The assembly code is as follows:

```

01 .MODEL SMALL
02 .STACK 100h
03
04 .DATA
05 WELCOME_MSG DB 'This is EXE program output!', 0Dh, 0Ah, '$'
06
07 .CODE
08 MAIN PROC
09 ; initialize DS
10 MOV AX, @DATA
11 MOV DS, AX
12
13 ; Display the string <INT 21h, Function 09h>
14 LEA DX, WELCOME_MSG
15 MOV AH, 09h
16 INT 21h
17
18 ; Exit Program <INT 21h, Function 4Ch>
19 MOV AH, 4Ch
20 INT 21h
21 MAIN ENDP
22
23 END MAIN
24
25
26
27

```

The registers pane shows the following values:

Register	Value
AX	07 20
BX	00 00
CX	01 30
DX	00 00
SI	0722
IP	0008
SS	0710
SP	0100
BP	0000
DI	0000
SI	0000
DS	0720
ES	0700

The assembly code pane shows the following instructions being executed:

```

07220: B8 184 ? MOV AX, 00720h
07221: 20 032 SPA MOV DS, AX
07222: 07 007 BEEP MOV DX, 00000h
07223: 8E 142 ? MOV AH, 09h
07224: D8 216 ? INT 021h
07225: B8 186 ? MOV AH, 04Ch
07226: 00 000 NULL INT 021h
07227: 00 000 NULL NOP
07228: B4 180 ? NOP
07229: 09 009 TAB NOP
0722A: CD 205 = NOP
0722B: 21 033 ? NOP
0722C: B4 180 ? NOP
0722D: 4C 076 L NOP
0722E: CD 205 = NOP
0722F: 21 033 ? NOP
07230: 90 144 E NOP
07231: 90 144 E NOP
07232: 90 144 E NOP
07233: 90 144 E NOP
07234: 90 144 E NOP
07235: 90 144 E NOP
07236: 90 144 E NOP
07237: 90 144 E ...

```

-
- The screenshot displays the 8086 emulator interface with the following components:
- Top Menu Bar:** File, math, debug, view, external, virtual devices, virtual drive, help.
 - Control Buttons:** Load, reload, step back, single step, run, and a step delay slider set to 0 ms.
 - Registers Panel:**
 - AX:** 09 24
 - BX:** 00 00
 - CX:** 01 30
 - DX:** 00 00
 - SI:** F400
 - IP:** 0204
 - SS:** 0710
 - SP:** 00FA
 - BP:** 0000
 - DI:** 0000
 - SI:** 0000
 - DS:** 0720
 - ES:** 0700
 - Memory Windows:**
 - F400:0204:** F4200: FF 255 RES, F4201: FF 255 RES, F4202: CD 205 =, F4203: 21 033 !, **F4204: CF 207 ±**, F4205: 00 000 NULL, F4206: 00 000 NULL, F4207: 00 000 NULL, F4208: 00 000 NULL, F4209: 00 000 NULL, F420A: 00 000 NULL, F420B: 00 000, F420C: 00 000, F420D: 00 000, F420E: 00 000, F420F: 00 000, F4210: 00 000, F4211: 00 000, F4212: 00 000, F4213: 00 000, F4214: 00 000, F4215: 00 000, F4216: 00 000, F4217: 00 000.
 - BIOS D1:** INT 021h, **IRET**, ADD [BX + SI], AL, ADD [BX + SI], AL, ADD [BX + SI], AL, ADD [BX + SI], AL, ADD [BX + SI], AL, ADD [BX + SI], AL, ADD [BX + SI], AL.
 - Assembly Code Window:**

```

01 .MODEL SMALL
02 .STACK 100h
03
04 .DATA
05 WELCOME_MSG DB 'This is EKE program output!', 0Dh, 0Ah, '$'
06
07 .CODE
08 MAIN PROC
09 ; Initialize DS
10 MOV AX, @DATA
11 MOV DS, AX
12
13 ; Display the string (INT 21h, Function 09h)
14 LEA DX, WELCOME_MSG
15 MOV AH, 09h
16 INT 21h
17
18 ; Exit Program (INT 21h, Function 4Ch)
19 MOV AH, 4Ch

```
 - Emulator Screen (80x25 chars):**

```

This is EKE program output!

```
 - Bottom Panel:** Includes buttons for 'clear screen' and 'change font', and a font size dropdown set to 8x16.

THE END